

Qatar-1b

R-2806

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140720 · created 2026-09-05T00:49:54+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456858.91441 +/- 0.00034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.057
Transit depth (Rp/Rs)^2	3.69 +/- 2.18 [%]
Orbital Inclination (inc)	83.06 +/- 2.85
Airmass coefficient 1 (a1)	1.07 +/- 0.16
Airmass coefficient 2 (a2)	-0.06 +/- 0.24
Scatter in the residuals of the lightcurve fit is	14.61 %
Best Comparison Star	#2 - [565, 154]
Optimal Method	PSF photometry
Transit Duration (day)	0.059 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.192 ± 0.057 vs 0.1463 (deviation 0.8σ)
Duration fit vs published	0.059 d vs 0.0692 d (deviation 0.41σ)
Mid-time O-C	0.8 min
Depth SNR	3.4
Residual scatter	14.61 %

Light curve

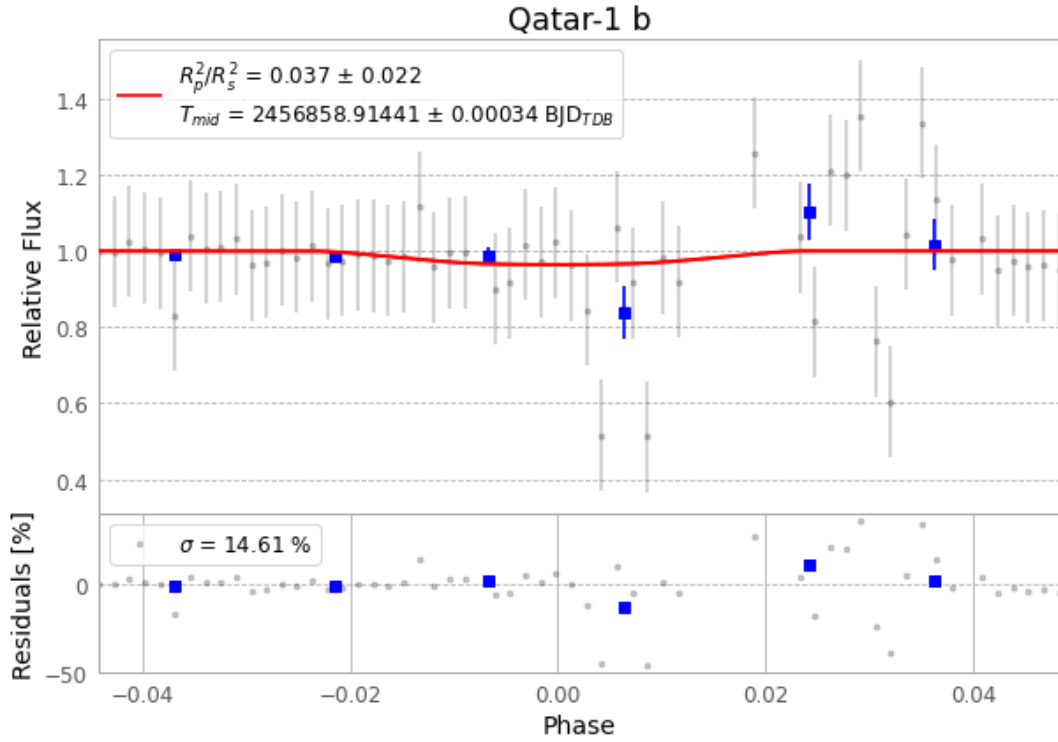


Figure 1 — FinalLightCurve_Qatar-1 b_2014-07-20.png (SHA-256 5f9bcef07655b0f5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 315], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f7d40f47b1949b33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), Qatar-1140720082412.FITS → Qatar-1140720113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140720.log (509dc0be800f...), FinalLightCurve_Qatar-1 b_2014-07-20.png (5f9bcef07655...), FinalLightCurve_Qatar-1 b_2014-07-20.csv (bee6d64aa6c6...)

Compute resources

Wall time 498.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 00:49Z.